MIL-B-36777A
31 August 1978
SUPERSEDING
MIL-B-36777
24 August 1971

MILITARY SPECIFICATION

BLOWTORCH, ALCOHOL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a hand-operated alcohol blowtorch suitable for use in taking impressions, waxing, setting up teeth, and other dental prosthetic procedures.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be of the elements specified and numerically listed in Figure 4 for the assembly component indicated. The plastic specified shall be a rigid, thermosetting, plastic material suitable for the intended purpose. When materials are used which are not specifically designated, they shall be of high quality throughout, and suitable in all respects for the intended purpose.

3.2 Design, style, and dimensions. The blowtorch assembly shall conform to Figures 1, 2, and 3 in design, style, and dimensions. The blowtorch shall be contoured to fit the operator's hand while the trigger is being operated. design of the blowtorch shall be such that tightening the thumbscrew valve shall seal its gasket and the accompanying operation of the trigger will produce a constant flow of air and a steadily projected, needlepoint flame. a slight turn or opening of the thumbscrew valve on the nozzle holder shall produce an intermittent or brush type flame suitable for spot-heating. the blowtorch shall be rendered suitable for use as an alcohol lamp by turning the nozzle holder out of the flame area. A rubber ring shall be friction-attached to the torch base for shock absorber and protective purposes.

3.3 Component parts. The blowtorch shall consist essentially of a top cover, filler cap, wick holder, nozzle holder, and a body which shall encircle or accommodate an alcohol reservoir, air pump, front plate, air reservoir, required air passages, and necessary screw and gasketing components numerically listed in Figure 4.

3.3.1 Top cover. The top cover shall be of a pear-shaped design and shall be provided with nine holes to permit its attachment to the blowtorch body. The cover shall also be furnished with two tapped holes to receive the threaded ends of the wick holder and the filler cap. The cover shall have a female metal insert embodied therein to receive the nozzle holder. The gasket, which isolates the air passage from the alcohol reservoir and the gasket, installed at top periphery of the torch body, shall be both compressed and sealed by the nine screws used to attach the cover to the body.

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3.3.2 Filler cap. The filler cap shall be threaded on one end to permit its being turned into or backed out of the top cover and its removal shall provide a port for filling the alcohol reservoir. A leather gasket shall be provided and kept in place on the threaded end of the cap to prevent alcohol leakage in the flame burning area.

3.3.3 Wick holder. The wick holder shall consist of a metallic holder and cap and a cotton wick. The holder shall be threaded on one end to permit its attachment to or removal from the top cover for wick replacement purposes. A leather gasket shall be provided and kept intact on the threaded end of the holder to insure a tight seal against alcohol leakage. The wick cap shall be attached to the nozzle holder by a metal bead chain. The cap shall be placed on the filler cap, for convenience, when the torch is in use and shall rest on the exposed section of the wick to lessen alcohol evaporation and wick incrustation, when not in operation. The wick shall be furnished in a size suitable for use in the holder and an extra wick shall be provided to serve as a spare part (see 3.7).

3.3.4 Nozzle holder. The nozzle holder shall consist essentially of a metallic male insert partially covered with heat-resisting molded plastic. The nozzle holder shall be furnished with a male thread on one end for attachment of a nozzle and a female thread on the opposite end for attachment of a thumbscrew valve. The uncovered, lower extension of the male insert shall be tapered to fit a correspondingly shaped metal female insert rigidly molded in the top cover. The tapered fit of these two parts shall afford a seal of the opposing metal surfaces and shall provide for rotation of the nozzle holder about the tapered axis. A metal shield shall be provided to protect the plastic portion of the nozzle holder from the heat of the nozzle flame and it shall be retained in place on the holder, along with the wick cap chain, by the nozzle being tightened on the holder.

3.3.5 Alcohol reservoir. The alcohol reservoir shall be located in the top section of the torch body and shall be of not less than 90 cubic centimeter (cc) capacity. The reservoir shall accommodate denatured alcohol base liquids of the type used to reduce shellac.

3.3.6 Air pump. Compressed air shall be forced into the air reservoir by a trigger-operated, air pump which shall consist of a piston, equipped with a groove-retained, synthetic rubber packing and tapped for attachment of the threaded end of a moving valve. The threaded end of the moving valve shall have a cross air passage for the air inlet adjacent to the piston and a threaded air passage drilled axially to intersect the first passage and a third air passage at the other end of the piston. The additional passage shall have a seat for the ball and spring which shall be held in position, as a check valve, by the counterbore of the piston. The check valve shall be closed on the pressure stroke of the piston and opened

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on the return stroke to recharge the pump cylinder with air. The moving valve shall have slots on its side to key in a corresponding female shape in the cylinder and shall have a spring inserted between the open end of the valve and the cylinder body to return the piston after its pressure stroke. A key slot, at the end of the moving valve, shall engage a corresponding male key in the trigger to act as an anti-rotation device when the molded plastic trigger is attached by screw.

3.3.7 Front plate. The front plate shall be interposed between the piston and trigger, shall be attached to the torch body by two screws, and shall be used to retain the piston and packing in the air cylinder of the torch body. The trigger shall contain a slot at its open end which shall align with the cross hole in the moving valve.

3.3.8 Air reservoir. An air reservoir having a volume of not less than 105 cc shall be located in the lower part of the blowtorch body and shall be connected to the pump cylinder by a cross hole. A fixed valve, equipped with a ball and spring, shall be interposed between the air reservoir and cross hole and shall serve as a check valve to prevent siphoning of the pressured air by the piston on its return stroke. One gasket shall seal the fixed valve and shall insure the complete isolation of the reservoir from the piston. The reservoir shall be encased by two gaskets and a bottom cover which shall be secured by a screw turned into the tapped insert of the blowtorch body. The head of this screw shall be concealed by an aluminum, spring-flanged cap which shall be pressed into position in the bottom cover of the torch. The air reservoir shall be connected to the female tapered insert (see 3.3.1 and 3.3.4) rigidly molded in the top cover by a vertical passage in the torch body.

3.4 Finish. All exposed metal parts, other than corrosion-resistant steel, shall be nickel-plated, except plating shall not be required on the brass nozzle, aluminum thumbscrew valve, or the brass threaded ends of the nozzle holder. Plastic shall be smoothly finished and polished.

3.5 Identification marking. The outer surface of the blowtorch bottom component shall be permanently and legibly marked with name or registered trademark of the manufacturer.

3.6 Instructions. Complete and detailed instructions for use of the blowtorch shall be furnished with each unit.

3.7 Spare parts. One extra wick shall be provided with each blowtorch.

3.8 Workmanship. Workmanship shall be first class throughout. The blowtorch shall be symmetrically made, smoothly and properly finished, and free from defects which detract from its appearance or impair its servicability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations required to assure conformance of those devices or lots of devices to be offered are classified as specified in tables I (see 4.2.1) and II (see 4.2.2). The tests required for acceptance are specified in table III (see 4.2.3).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL) shall be as indicated in table I. The unit of product for examination purposes shall be one blowtorch.

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TABLE I. Inspection and acceptable quality levels.

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Examination. The devices shall be examined for the defects listed in table II. Examination shall not be restricted to the classified possible defects listed below.

TABLE III. Classification of defects.

Categories	Defects
Major A:	
101	Wick replacement parts missing from assemblies.
102	Sharp edges, nicks, and scratches present on torch assembly.
103	Torch components not properly assembled.
104	Plating shows signs of peeling or flaking.
105	Design not as specified.
106	Valve packing not as specified.
Major B:	
151	Torch body has poorly molded surface finish.
152	Taper of nozzle inserts not as specified.
153	Nozzle shield shows signs of corrosion.
154	Gaskets missing on wick holder and filler cap components.
Minor A:	
201	Finish of wick holder and filler components not as specified.
202	Identification marking incorrect, incomplete, or illegible.
Minor B:	
251	Dimensions not as specified.
252	Set of instructions missing from assembly.

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4.2.3 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III, except that the acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be one blowtorch.

TABLE III. Sampling and quality conformance criteria for testing.

Characteristic	Requirement	Test procedure	Inspection level
Performance	3.2	4.4.1	S-2

4.3 Packaging inspection. The inspection of the unit package, packing, and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to detemine compliance with the requirements of this specification.

4.4.1 Performance test. Performance tests shall be performed on the blowtorch assembly to determine that it is capable of meeting the requirements imposed on it by the dental prosthetic procedures outlined in the Scope section (see 1.1) of this specification. These tests shall be preceded by filling the alcohol reservoir with the prescribed type of liquid and checking, for safety reasons, that the cap filler and wick holder gaskets are held tightly in place and that there is no leakage around these two ports in the top cover of the blowtorch. The assembly shall then be subjected to the two tests, outlined in section 3.2, that will successively check the unit's capability to produce: (1) A steadily projected, needlepoint flame for burning purposes or (2) An intermittent, or brushing flame suitable for spot-heating procedures. The flexibility of the male-female tapered joint inserts, described in section 3.4, shall be checked by alternately rotating the nozzle holder to the right and left of the wick flame area which shall be a nozzle positioning capability required in converting the torch to alcohol lamp usage.

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5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One blowtorch, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit packaging. Each torch shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. The box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against damage to the contents. Closure shall be as specified in the applicable box specification.

NOTE: Blowtorch shall not contain alcohol or alcohol vapors when packaged.

5.1.3 Level C. Each torch shall be packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Sixty unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the box specification.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of the box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Sixty unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

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5.2.3 Level C. The packaged torches shall be packed in exterior containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. The contract or purchase order number and name of contractor shall also be shown.

5.3.2 Exterior container. Exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Limitations. This specification does not include all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Federal Government requirements.

6.3 This specification covers the following item appearing in the Federal Supply Catalog:

NSN	Item Identification
6520-00-561-7150	BLOWTORCH, ALCOHOL

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

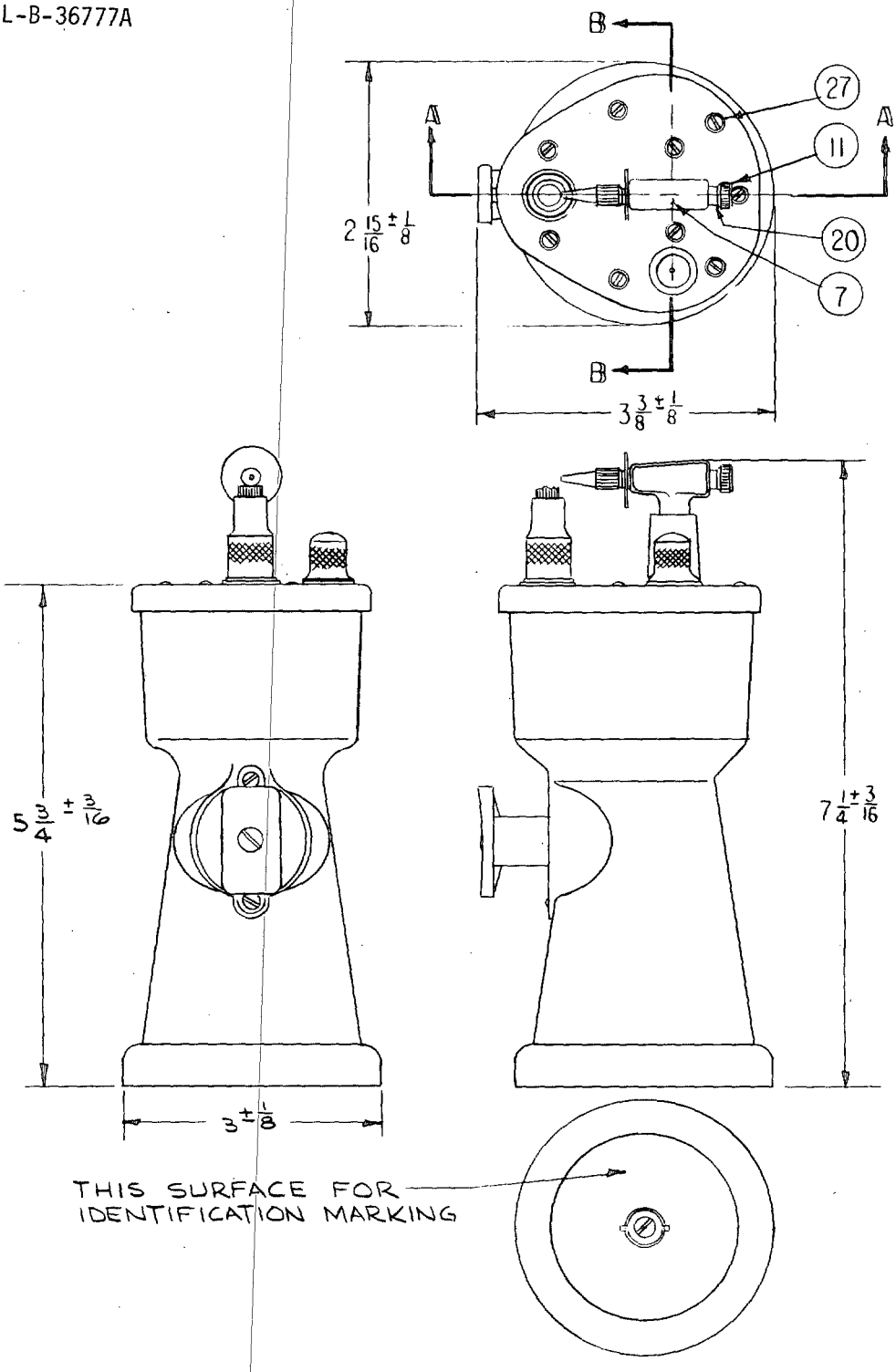
Custodians:
Army - MD
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Preparing activity:
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BLOW TORCH, ALCOHOL

FIGURE 1

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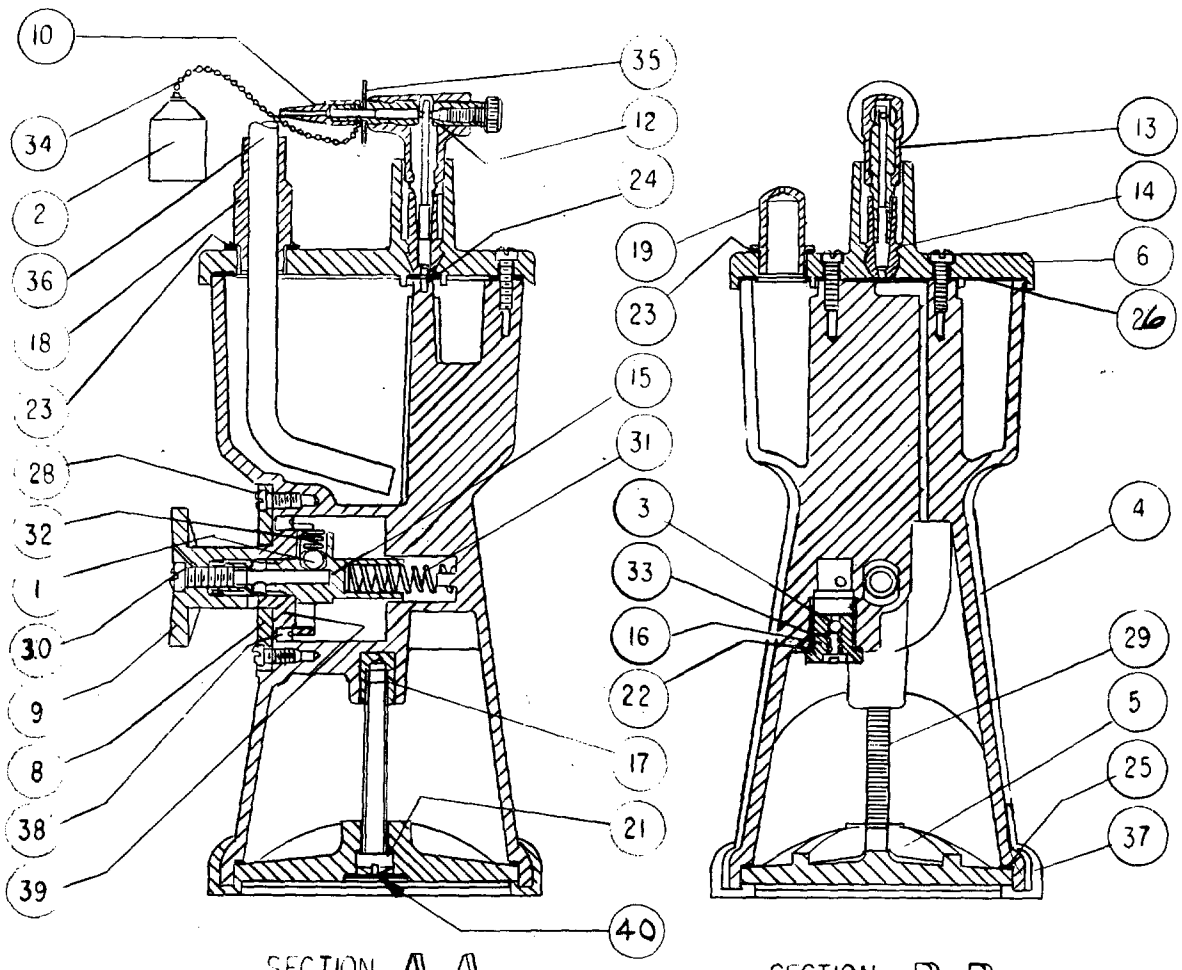


FIGURE 2

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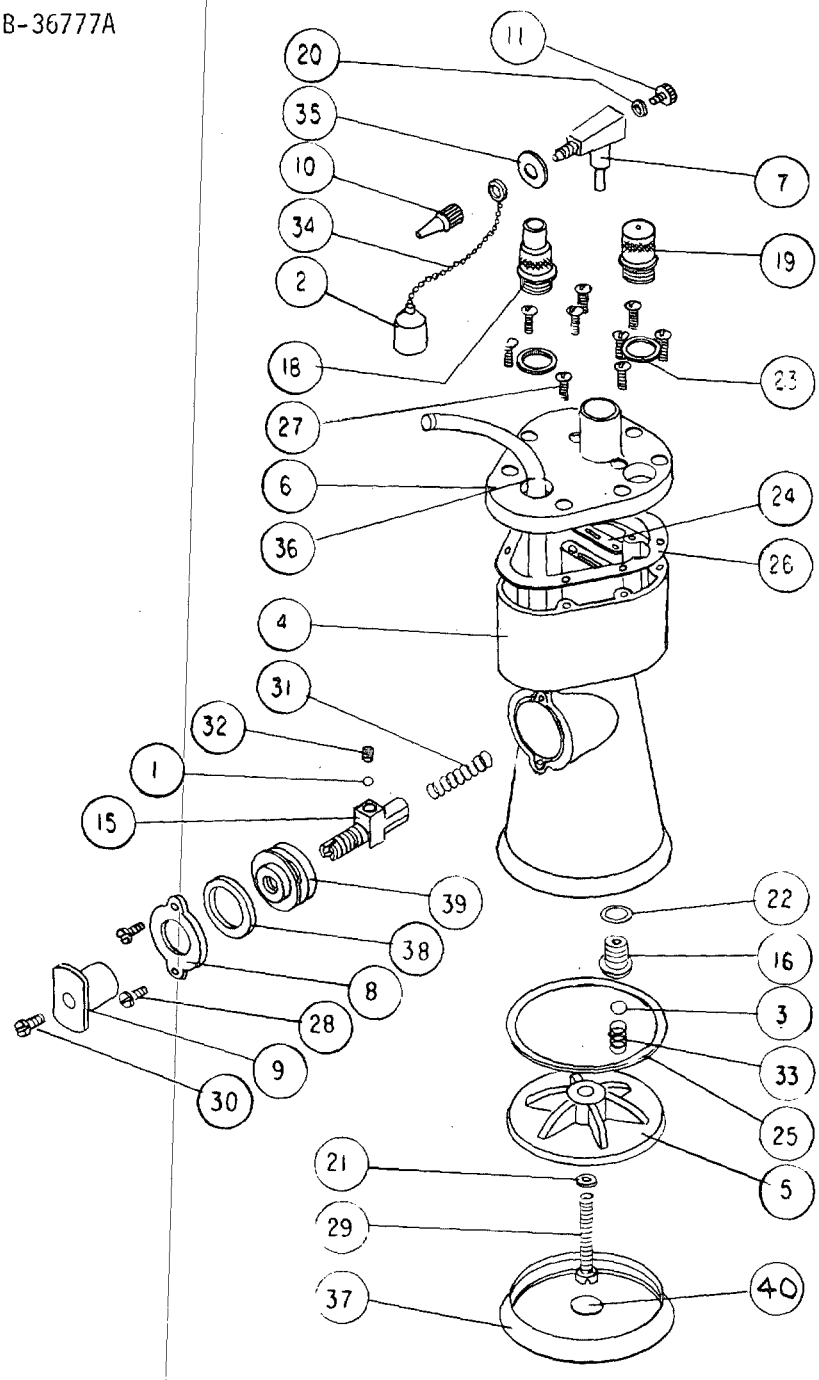


FIGURE 3

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BLOWTORCH, ALCOHOL, 6520-561-7150

PART NO.	PART NAME	MATERIAL	QTY. REQ'D
1	BALL, MOVING VALVE	BRASS	1
2	WICK CAP	FREE CUTTING YELLOW BRASS	1
3	BALL, FIXED VALVE	BRASS	1
4	BODY	BLACK PHENOLIC	1
5	BOTTOM	BLACK PHENOLIC	1
6	COVER, TOP	BLACK PHENOLIC	1
7	NOZZLE HOLDER	BLACK PHENOLIC	1
8	FRONT PLATE	BLACK PHENOLIC	1
9	TRIGGER	BLACK PHENOLIC	1
10	NOZZLE	FREE CUTTING YELLOW BRASS	1
11	THUMBSCREW VALVE	ALUMINUM	1
12	NOZZLE MOUNTING INSERT	FREE CUTTING BRASS	1
13	MALE TAPERED INSERT	COLD FINISHED CARBON STEEL	1
14	FEMALE TAPERED INSERT	FREE CUTTING YELLOW BRASS	1
15	MOVING VALVE	ZINC	1
16	FIXED VALVE	FREE CUTTING YELLOW BRASS	1
17	TAPPED INSERT, AIR RESERV.	FREE CUTTING YELLOW BRASS	1
18	WICK HOLDER	FREE CUTTING YELLOW BRASS	1
19	FILLER CAP	FREE CUTTING YELLOW BRASS	1
20	GASKET, THUMBSCREW	FIBRE COMPOSITION	1
21	GASKET, SCREW	FIBRE COMPOSITION	1
22	GASKET, FIXED VALVE	FIBRE COMPOSITION	1
23	GASKET, FILLER & WICK	LEATHER	2
24	GASKET, FEMALE INSERT	SYNTHETIC RUBBER	1
25	GASKET, BOTTOM	FIBRE	1
26	GASKET, TOP COVER	SYNTHETIC RUBBER	1
27	SCREW, TOP COVER	STEEL, COMMON FASTENER	9
28	SCREW FRONT PLATE	BRASS, COMMON FASTENER	2
29	SCREW, BOTTOM	BRASS, COMMON FASTENER	1
30	SCREW, TRIGGER	BRASS, COMMON FASTENER	1
31	SPRING, MOVING VALVE	MUSIC WIRE	1
32	SPRING MVG. VALVE CHECK	PHOSPHOR BRONZE	1
33	SPRING FIXED VALVE CHECK	PHOSPHOR BRONZE	1
34	CHAIN, WICK CAP	BRASS	1
35	SHIELD, NOZZLE	COLD ROLLED STEEL	1
36	WICK	COTTON	2
37	RING, BOTTOM	SYNTHETIC RUBBER	1
38	PACKING, MOVING VALVE	SYNTHETIC RUBBER	1
39	PISTON, MOVING VALVE	ALUMINUM	1
40	HANAU MONOGRAM	ALUMINUM	1

FIGURE 4

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-B-36777A BLOWTORCH, ALCOHOL

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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